

Jiaming Shan

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Profile

Computer Science researcher in UCSB's direct-entry Ph.D. program, pursuing an M.S. expected July 2027 and available for full-time work July 15, 2027. Focused on efficient language-model architectures, agent evaluation, and post-training from real user feedback at Tencent Cloud WorkBuddy.

Education

University of California, Santa Barbara (UCSB)

Direct-entry Ph.D. program; M.S. in Computer Science expected

Sep 2024 - Jul 2027 (expected)

Santa Barbara, CA

- Research focus: LLM efficiency, adaptive and long-context attention, and autonomous agents.
- GPA: **3.95/4.00**; advisor: Prof. Xifeng Yan; coursework in the top 5% of the program.

Shanghai Jiao Tong University

B.S. in Computer Science, ACM Honors Class

Sep 2020 - Jun 2024

Shanghai, China

- Selected for an honors program for the top 5% of computer science students; core-course GPA: **3.98/4.3**; rank: **3/36**.
- 2023 National Scholarship; Zhiyuan Honorary Scholarship, 2020 - 2023.

Research Experience

University of California, Santa Barbara (UCSB)

Graduate Researcher - LLM Systems and Agents

Sep 2024 - Present

Santa Barbara, CA

- Co-first-authored **Learning When Not to Attend Globally (All-or-Here Attention, AHA)** (manuscript): designed a token-level binary router that switches each attention head between global and sliding-window attention.
- Implemented inference-speedup paths and experiments in PyTorch, Triton, and FlashAttention. On OLMo-2-1B with a 128-token window, **88.4%** of token-head decisions chose local attention with comparable aggregate benchmark performance; AHA also improves DuoAttention as an orthogonal routing layer.
- Built benchmark and environment abstractions for **Proactive Agent Research Environment**, including finite-state-machine app models, goal inference, and multi-app orchestration evaluation.

Massachusetts Institute of Technology (MIT)

Summer Research, CSAIL

Aug 2023 - Jun 2024

Boston, MA

- Implemented physically constrained agents and helper bots in ThreeDWorld and helped build an LLM-based cooperative-agent benchmark published at **NeurIPS 2024** (Datasets and Benchmarks Track).

MIT / Shanghai Jiao Tong University

Summer Research (Remote)

Feb 2023 - May 2023

Remote

- Co-developed a framework that converts environment states into text prompts for cooperation; contributed to VirtualHome and heuristic-baseline implementations and conducted user studies comparing LLM agents with classical planning methods for an **ICLR 2024** paper.

Internship Experience

Tencent Cloud WorkBuddy

Large Language Model Agent Research Intern

Sep 2026 - Present

- Used user feedback to locate failed actions in agent trajectories; conditioned stronger-model rerollouts on user hints to synthesize repaired continuations for post-training.
- Observed that fine-tuning on hint-conditioned traces improved task behavior but increased tool calls; proposed mixing high-quality post-training data to address a possible distribution shift.

Shanghai Yuanwo Intelligent Technology Co., Ltd. (RWKV)

Agent Engineering Intern

Mar 2026 - Jun 2026

Remote

- Built document summarization and retrieval pipelines around local, high-concurrency RWKV deployments.
- Added tool-output summarization and key-information extraction to reduce downstream context usage; generated HTML result pages concurrently to improve readability.

Riema Labs

Research Intern

Mar 2025 - Jul 2025

- Contributed to MathEye research on automated repair for LLM-generated Lean proofs, based on *From Sorry to QED: Refining LLM Math Proofs via Angelic Execution*; used angelic execution to abstract failed fragments, proof dependency graphs and MaxSAT to select key obligations, and automatically reconstructed verifiable proofs.

Selected Papers and Manuscripts

- **Learning When Not to Attend Globally** (AHA). Xuan Luo*, Jiaming Shan*, Wesley Truong, Kailai Zhang, Hanzhe Zhang, Xifeng Yan. Manuscript. (* Equal contribution.)
- **Proactive Agent Research Environment: Simulating Active Users to Evaluate Proactive Assistants**. Deepak Nathani, Cheng Zhang, Chang Huan, Jiaming Shan, et al. arXiv preprint.
- **Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge**. Co-author, NeurIPS 2024, Datasets and Benchmarks Track.
- **Building Cooperative Embodied Agents Modularly with Large Language Models**. Co-author, ICLR 2024.

Selected Engineering Projects

Sentinel-Rosette: [GitHub](#). Independently built a symbolic EVM in Rosette, integrated Foundry execution traces, and used MaxSMT to search exploit parameters; packaged the prototype in Docker.

Claude Link: [GitHub](#). Built an MCP/tmux prototype for persistent, observable Claude Code sessions with message routing and human intervention across collaborating agents.

Honors and Awards

- 2023 National Scholarship Award.
- 2020 - 2023 Zhiyuan Honorary Scholarship, Shanghai Jiao Tong University (top 2%).
- 2021 Interdisciplinary Contest in Modeling, Honorable Mention.
- 2021 National Undergraduate Mathematical Contest in Modeling, Provincial Third Prize.
- 2021 International Physics Competition for University Students, Silver Medal.

Technical Skills

Programming Languages:	Python, C++, Racket/Rosette, Solidity
AI / Systems:	PyTorch, Hugging Face, Triton, FlashAttention, vLLM, Weights & Biases
Agent Workflows:	MCP, Claude Code, Codex, OpenClaw, Cursor, Tmux
Tools / Deployment:	Git, Docker

Links

[Personal Website](#) [Google Scholar](#) [GitHub](#) [Zhihu](#) [LinkedIn](#)